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ORIGINAL CONTRIBUTION

Topological Synthesis of Planar Kinematic and Epicyclic Geared Mechanisms by Application of Total Distance Technique

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Abstract
Topological synthesis and isomorphism detection of planer kinematic mechanisms (PKM) and epicyclic geared mechanisms (EGM) have been major issues to avoid duplicity since last several years. In this paper, an innovative approach based on Total distance technique has suggested to identifying the isomorphic properties of different mechanisms. This technique presents quantitative method by application of graph theory and adjacency matrix. Finally, proposed technique is examined by using different examples of different mobility and various number of links of planar kinematic and epicyclic geared mechanisms. All results are completely satisfied with references taken from the literature.

Keywords Planar kinematic chain · Isomorphism · Mechanism · Total distance technique · Graph theory

Introduction

Synthesis and generation of planar kinematic mechanisms (PKMs) and epicyclic geared mechanisms (EGMs) with higher number of links and degree of freedom is a major research area by several authors. Although many researchers have reported different approaches to detect isomorphic property and generated new graphs for PKMs and EGMs, but each has its own limitations. Nowadays, evaluation of isomorphic properties in generation of EGMs with higher link and degree of freedom (DOF) is the key requirements to avoid the duplicity in mechanism, synthesis & analysis, and kinematic simulation of EGTs. In archival literature several studies related to modeling of epicyclic gear trains have been reported by various researchers as presented here. F. Freudenstein [1] pioneered the structural classification of mechanisms, introducing Polya's theory and enumerating eleven mechanisms from a four-bar kinematic chain and its inversions. L. Dobrjanskyj and F. Freudenstein [2] furthered this by employing graph theory for structural analysis, offering insights into mechanical structures, automatic sketching, and systematic classifications. Z. Levai [3] simplified structural analysis and synthesis of EGTs, deriving 34 arrangements from a basic EGT and providing notation for gear meshing. Buchsbaum and Freudenstein [4] extended this by applying network theorems for systematic enumeration of gear transmission and differential gearing. Freudenstein [5] developed a Boolean algebra-based algorithm for kinematic structure investigation, aiding in dynamic analysis, computer-aided sketching, and animation. Jiyaul et al. [6] utilized signal flow graphs for kinematic analysis of planetary gear trains (PGTs), determining angular velocity and transmission ratios. Uicker Jr and Raicu [7] introduced a methodology for identifying kinematic chains and detecting isomorphism, while Mrutyunjaya and Raghavan [8] devised a methodology using Bocher's formulae for isomorphism evaluation in kinematic chains. Allen [9] proposed a bond graph model for kinematic and dynamic analysis of gear drive transmission systems, enabling determination of

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torque transmission equations and velocity ratios. Day et al. [10] employed a tabulation method for analyzing speed ratios of PGTs, aiding designers in synthesizing correct speed ratios. Gibson and Kramer [11] introduced a symbolic notation for describing two-DOF spur planetary gear trains (SPGTs) and presented 22 basic EGT configurations for kinematic equations determination. Ravisankar and Mruthyunjaya [12] developed a computerized methodology for structural analysis and synthesis of kinematic chains. A novel method [13] utilizing Wiener number for isomorphism detection in epicyclic geared mechanisms and planar kinematic chains resolved by Mustafa & Hasan for long-standing duplicity issues. Jiyaul et al. [14] have outlined a graph theory-based approach for non-duplicative generation of one-degree-of-freedom epicyclic gear trains, employing adjacency matrices and Wiener numbers.

Peng et al. [15] proposed a variable structure approach for calculating speed ratios in PGTs with variable degrees of freedom, offering flexibility in design. Pennestri et al. [16] developed algorithms based on graph theory for dynamic and static analysis of geared mechanisms, facilitating automatic generation and calculation processes. Chen and Chen [17] introduced a stable model for efficiency calculation in EGTs, closely aligning theoretical values with experimental data. Ding et al. [18] synthesized 1-DOF EGTs up to nine links using graph representation and characteristic number strings for isomorphism identification. Saini and Singh [19] and Rajasri et al. [20] employed hamming number techniques for isomorphism detection in KCs and EGTs, streamlining graphical analysis and synthesis processes. Mustafa et al. [21] have compared modified path matrix, Modified gradient, and Bocher's technique for isomorphism in epicyclic Gear Trains, assessing reliability, computational efficiency, and structural properties. Mustafa et al. [22] introduced a modified path matrix approach for isomorphism detection in epicyclic gear trains, addressing challenges in existing methods, validated through examples. Chu and Zou [23] provided insights into structural synthesis and topological graphs of planar multiple joint and geared kinematic chains. Yang et al. [24] proposed a novel approach for detecting rotational graphs and canonical rotational graphs of PGTs, introducing concepts of graph representation to avoid pseudo-isomorphism. Gao and Hu [25] delved into the topological synthesis and kinematic analysis of planetary transmission systems using graph theory, highlighting the relationship between speed ratios based on topology [26]. Kamesh et al. [27] introduced parameters for the synthesis of EGTs, enabling the detection of displacement and rotational graphs. Innovative Modified Gradient Method (MGM) proposed by Mustafa et al. [28] for isomorphism detection in epicyclic gear trains, providing a novel, efficient algorithm validated by application.

Meanwhile, Alam et al. [29] given idea about machine learning technique for EGTs. Esmail [30] conducted a comprehensive kinematic analysis of PGTs using graph theory, innovatively applying it to determine speed ratios in multiple link mechanisms [31]. Shukla et al. [32] employed the gradient method to develop an algorithm for isomorphism checking in KCs, introducing the concept of gradient strings for synthesis and analysis [33]. Rao et al. [34] introduced fuzzy entropy for rating epicyclic gear trains, providing insights for diverse machinery systems. Yang and Ding [35] devised an automated perimeter loop method for isomorphism detection in PGTs, streamlining duplicity avoidance. Mustafa et al. [36] introduced an innovative approach utilizing Bocher's equation for isomorphism identification in epicyclic gear trains. Liu et al. [37] proposed using a GA model for isomorphism identification in planar kinematic mechanisms, enhancing efficiency. Yang and Ding [38] developed a fully automatic method for detecting degenerate structures in EGTs, crucial for vehicle transmission systems and aeronautics applications [39, 40]. Machine learning technique has been suggested in papers [41–43] for correct evaluation and more data dependent method for structural synthesis of geared mechanisms and robotic mechanisms. A metamaterial concept on pipe [44] and rail track [45] is suggested by Iqbal et al., which can be implemented in robotic mechanism and parallel manipulator to control the vibration for robust design and accurate results. Table 1 is represented a systematic comparison between the proposed Total Distance Technique (TDT) and existing isomorphism detection methods, including the Modified Path Matrix, Gradient Method, Hamming Number Method, Bocher's Technique, Wiener Number Method, and Genetic Algorithm-based approaches. The comparison highlights their computational efficiency, accuracy, advantages, and limitations, demonstrating that the proposed TDT offers a simple, computationally efficient, and reliable approach for isomorphism detection in planar kinematic and epicyclic geared mechanisms while producing results consistent with established methods.

In this paper, a new method based on "Total distance technique" is introduced for identification of isomorphism in PKMs an EGMs. The proposed technique has been evaluated based on graph theory and adjacency matrix. A MATLAB coding is written for determination of total distance of adjacency matrix. If total distance of two PKM graphs or EGM graphs are equal, then they are isomorphic in nature otherwise they are not isomorphic in nature.

Table 1 Comparison of existing isomorphism detection methods with the proposed method

Method	Principle	Computational Efficiency	Accuracy	Advantages	Limitations
Bocher's Technique [8, 36]	Uses Bocher's equations and graph parameters for identifying isomorphism.	Moderate	High	Well-established and reliable for small mechanisms.	Mathematical formulation becomes cumbersome for mechanisms with higher numbers of links and complex topologies.
Modified Path Matrix (MPM) [21, 22]	Compares path matrices derived from graph connectivity.	Moderate	High	Capable of distinguishing structurally similar mechanisms with good reliability.	Requires construction and comparison of large path matrices, increasing computational effort for complex graphs.
Modified Gradient Method (MGM) [21, 28, 32]	Employs gradient values generated from graph topology.	Moderate	High	Provides effective discrimination among many planar and geared mechanisms.	Gradient calculations become increasingly complicated for large-scale mechanisms.
Hamming Number Method [19, 20]	Utilizes Hamming numbers derived from adjacency relationships.	Moderate	High	Simple numerical comparison and suitable for many kinematic chains.	Less effective when different graphs produce similar numerical characteristics; additional verification may be required.
Wiener Number Method [13]	Uses graph distance-based Wiener indices as graph invariants.	Moderate	High	Straightforward implementation and effective for many PKMs and EGMs.	Different non-isomorphic graphs may occasionally possess identical Wiener indices, requiring supplementary validation.
Genetic Algorithm (GA)-based Method [37]	Applies evolutionary optimization for graph enumeration and isomorphism detection.	Low (higher computational cost)	High	Suitable for complex optimization and automated mechanism generation.	Computationally intensive and dependent on algorithm parameters; longer execution time.
Net Distance Technique [26]	Computes net distance values from graph topology.	High	High	Simple implementation and reduced computational effort.	Primarily validated for limited classes of mechanisms and may require additional graph descriptors for highly complex structures.
Proposed Method (TDT)	Computes the total distance directly from the adjacency matrix by summing row and column elements.	Very High	High	Very simple algorithm, easy MATLAB implementation, low computational cost, suitable for both planar kinematic and epicyclic geared mechanisms, and scalable to mechanisms with higher numbers of links.	Validation in the present work is limited to planar kinematic and epicyclic geared mechanisms; future work will extend the approach to spatial mechanisms and larger graph databases.

Graph Theory Application

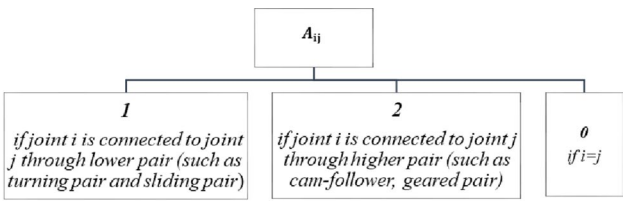
Graph theory is essential for detecting isomorphism and performing topological analysis in Parallel Kinematic Mechanisms (PKM) and Epicyclic Gear Mechanisms (EGM). It enables systematic identification of structurally identical configurations, optimizes design efficiency, and ensures kinematic integrity by mapping linkages, joints, and mobility constraints through adjacency matrices and graph-based algorithms.

Adjacency Matrix [A]

The adjacency matrix shows the connection between two vertices or links in PKMs and EGMs. The matrix demonstrates symmetry while all its diagonal positions contain zeros. The matrix derives from its functional schematic and shows as:

$$[A] = \{a_{ij}\} \tag{1}$$

The formula uses (i, j) to indicate the row-column positions of the elements from matrix [A]. A process has been established to establish each element value in the matrix according to the provided below chart Fig 1.



For n×n connectivity matrix can be written as,

$$[\mathbf{A}] = \begin{bmatrix} m_{11} & m_{12} & m_{13} & \cdots & \cdots & m_{1n} \\ m_{21} & m_{22} & m_{23} & \cdots & \cdots & m_{2n} \\ m_{31} & m_{32} & m_{33} & \cdots & \cdots & m_{3n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ m_{n1} & m_{n2} & m_{n3} & \cdots & \cdots & m_{nn} \end{bmatrix} \tag{2}$$

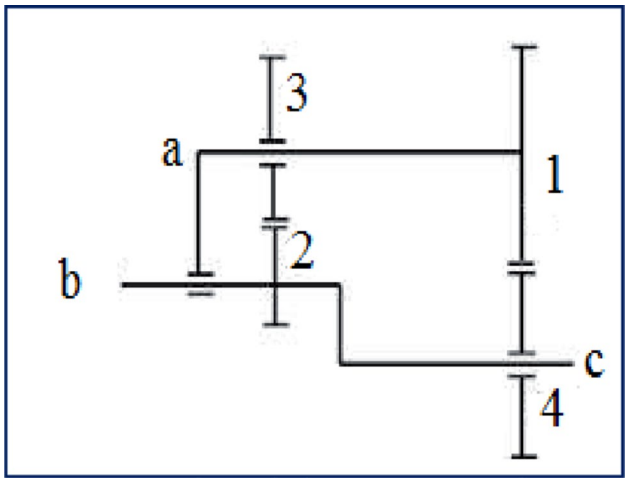


Fig. 1 Functional Schematic of four-links One-DOF EGMs

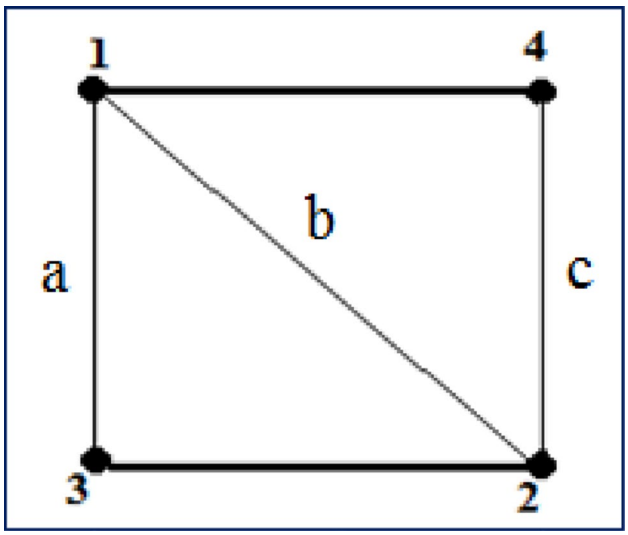


Fig. 2 Graph of four-links One-DOF EGMs (G-4001)

The adjacency matrix for four-link 1-DOF EGMs (Fig. 2) takes its form based on the given chart.

$$\text{Adjacency Matrix, } [A_m] = \{a_{4\times 4}\} = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 3 \\ 2 & 1 & 3 & 0 \end{bmatrix} \quad (3)$$

Total Distance Technique

The "Total distance technique (TDT)" is topological approach to evaluate isomorphic property in the mechanisms. The proposed method is evaluated using graph theory and the adjacency matrix. A computer program is developed to determine the total distance of the adjacency matrix by adding all rows and columns elements. If the values of TDT of two PKM or EGM graphs are equal, they are isomorphic; otherwise, they are not isomorphic.

Table 2 Total distance technique (TDT) for two EGMs with 4-links and single DOF

Fig. No.	Adjacency Matrix [A]	Total Distance (TDT)
Figure 3	$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 \end{bmatrix}$	10
Figure 4	$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 3 \\ 2 & 1 & 3 & 0 \end{bmatrix}$	20

$$\text{TDT} = \sum_{i=1}^{n}\sum_{j=1}^{n}(A) \quad (4)$$

Where A={a_{ij}}= Adjacency Matrix of a perticular conventional graph.

For Fig. 2, Value of TDT by using Eq. (4),

$$\text{TDT} = \begin{bmatrix} 4 \\ 4 \\ 6 \\ 6 \end{bmatrix} = 20$$

TDT shows equality, yet distinct mechanisms exist whenever the TDT differ.

Applications of Total Distance Technique (TDT)

The proposed method undergoes testing for EGMs with four, five and six links and six-link PKMs having 1–2 degrees of freedom. The article uses various examples to illustrate both the practical deployment of graph theory and its utilization of TDT.

Example 1 ***Total Distance Technique in two EGMs with 4-links and single DOF***.

Table 2 shows the values of TDT of these EGMs as different which proves Figs. 3 and 4 represent unique structural arrangements.

Example 2 ***TDT in two EGMs with five-links and single DOF in Fig. 5 and Fig. 6.***

Table 3 shows the values of TDT of these EGMs as different which proves Figs. 5 and 6 represent unique structural arrangements.

Example 3 ***TDT in two EGMs with Six-links and Two-degree of freedom in Fig. 7 and Fig. 8.***

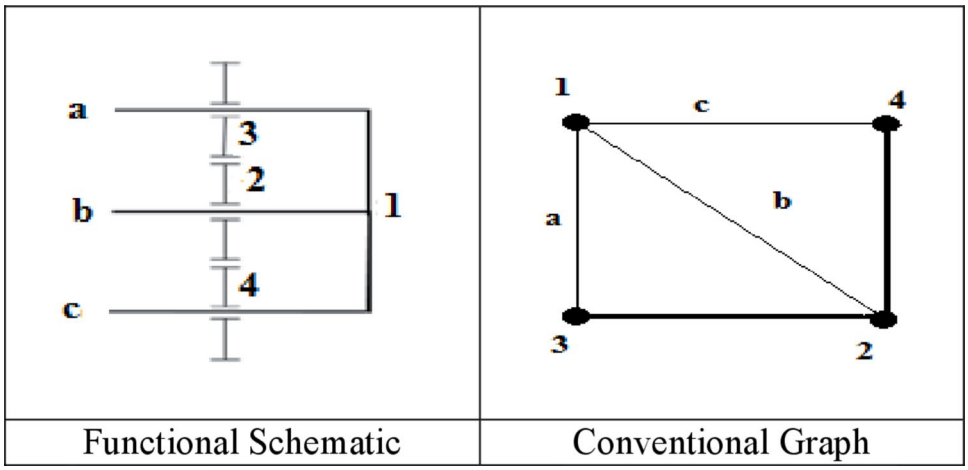


Fig. 3 Graph of 4-links 1-DOF EGM-4001

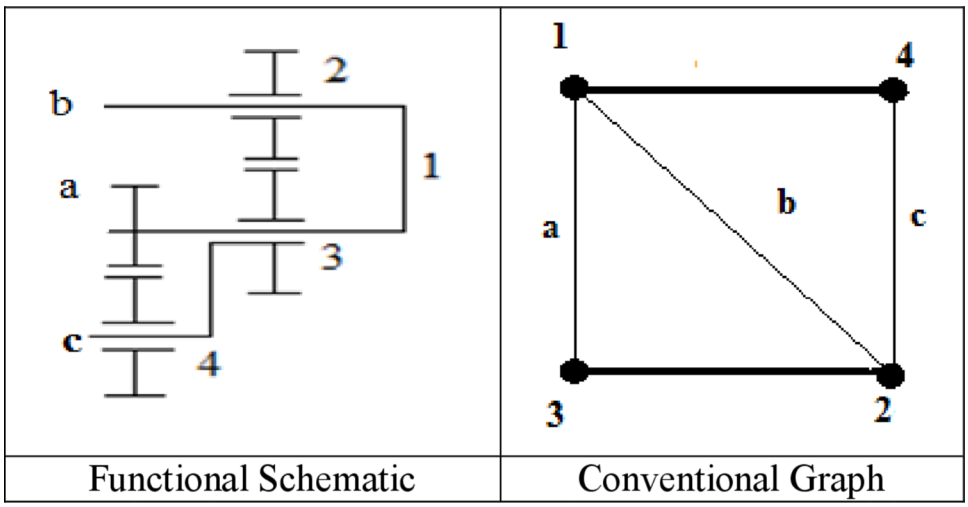


Fig. 4 Graph of 4-links 1-DOF EGM-4002

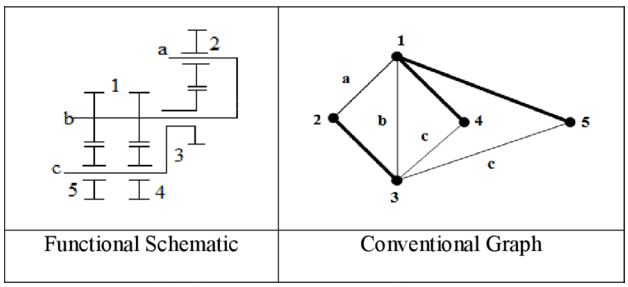


Fig. 5 Graph of 5-links 1-DOF EGM-5001

The TDT for these two EGMs show dissimilarity thus proving the graphs are distinct from one another based on Table 4.

Example 4 ***TDT in two PKMs with Six-links and One-degree of freedom***.

The different TDT between these two PKMs demonstrates that Figs. 9 and 10 represent distinct graphs according to Table 5. This example of planner kinematic chains is given for validating the proposed algorithm of Total distance technique for topological synthesis in PKMs and EGMs.

Example 5 ***TDT in two PKMs with Eight-links and One-degree of freedom***.

The different TDT between these two PKMs demonstrates that Figs. 11 and 12 represent distinct graphs according to Table 6. So, these two graphs of kinematic chains are non-isomorphism to each other.

Discussion

The proposed Total Distance Technique provides a simple and computationally efficient graph-theoretic approach for identifying isomorphism in planar kinematic mechanisms and epicyclic geared mechanisms. Unlike several existing methods that require complex graph invariants, characteristic polynomials, gradient strings, or iterative optimization procedures, the proposed method evaluates isomorphic

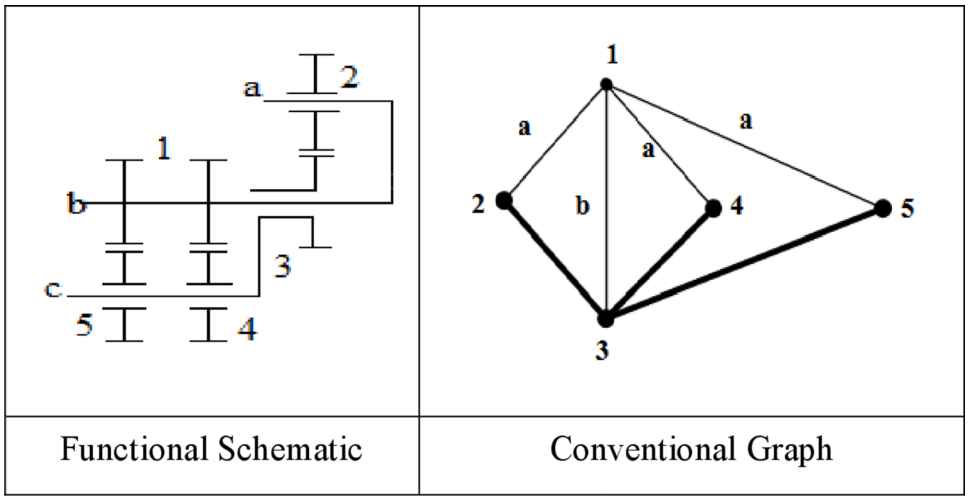


Fig. 6 Graph of 5-links 1-DOF EGM-5002

Table 3 TDT for two EGTs with five-links and single degree of freedom

Fig. No.	Adjacency Matrix [A]	Total Distance (TDT)
Fig 5	$\begin{bmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 3 & 1 & 0 & 2 \\ 2 & 3 & 1 & 2 & 0 \end{bmatrix}$	36
Fig 6	$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 2 & 2 & 0 \end{bmatrix}$	32

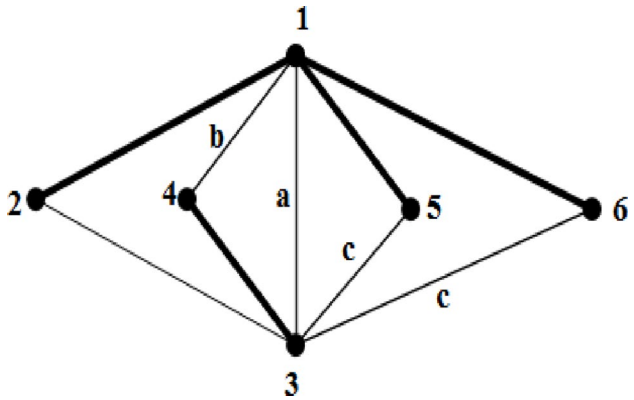


Fig. 7 Graph of 6-links EGM-6001

properties directly from the adjacency matrix by computing the total distance. This significantly reduces computational complexity while maintaining reliable discrimination between structurally distinct mechanisms.

The effectiveness of the proposed technique has been demonstrated through five representative case studies involving four-, five-, six-, and eight-link mechanisms with one and two degrees of freedom. In all investigated examples, different TDT values correctly distinguished non-isomorphic graphs, and the obtained results were fully consistent with previously published methods reported in the literature [14, 17, 21, 23], and [37]. These findings confirm the correctness and robustness of the proposed approach for topological synthesis and isomorphism detection.

From a practical perspective, the simplicity of the algorithm makes it attractive for implementation in computer-aided mechanism synthesis and automated graph generation. Since the method only requires the adjacency matrix and

Table 4 TDT for two EGMs with Six-links and Two-degree of freedom

Fig. No.	Adjacency Matrix [A]	Total Distance (TDT)
Figure 7	$\begin{bmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 3 & 3 \\ 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 1 \\ 2 & 3 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 1 & 2 & 0 \end{bmatrix}$	54
Figure 8	$\begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 & 3 & 3 \\ 2 & 2 & 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 3 & 3 & 1 & 2 & 3 & 0 \end{bmatrix}$	58

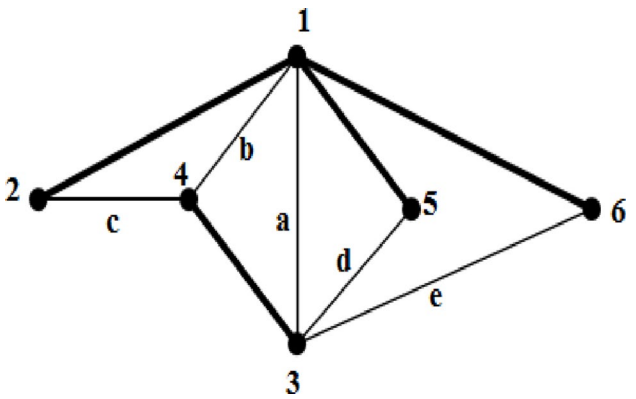


Fig. 8 Graph of 6-links EGM-6002

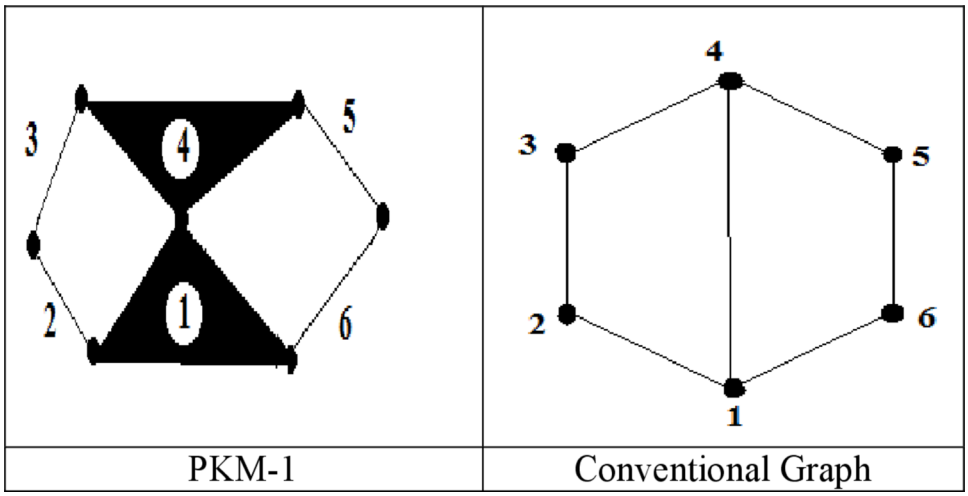


Fig. 9 Graph of Six-links One-DOF PKM-1

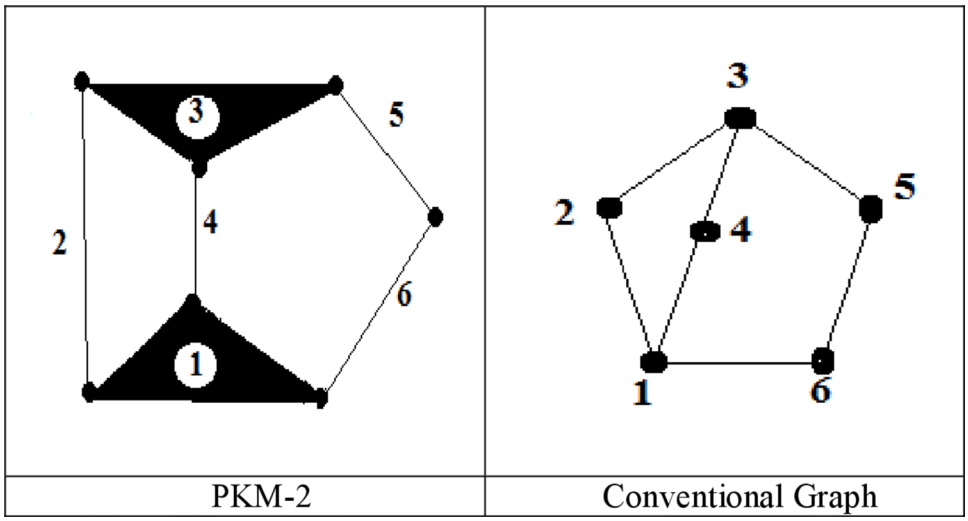


Fig. 10 Graph of Six-links One-DOF PKC-2

Table 5 TDT for two PKCs with Six-links and One-degree of freedom

Fig. No.	Adjacency Matrix [A]	Total Distance (TDT)
Figure 9	$\begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 2 & 1 & 0 & 1 & 2 & 3 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 2 & 3 & 2 & 1 & 0 & 1 \\ 1 & 2 & 3 & 2 & 1 & 0 \end{bmatrix}$	50
Figure 10	$\begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 2 & 1 & 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 2 & 1 & 0 \end{bmatrix}$	48

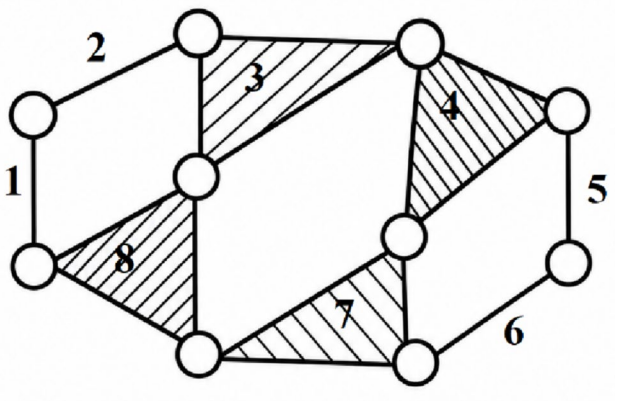


Fig. 11 Graph of Eight-links One-DOF PKC-1

basic matrix operations, it can be efficiently implemented in MATLAB or other programming environments with minimal computational effort. Consequently, the proposed TDT has the potential to serve as a fast-screening tool during the enumeration and synthesis of large numbers of candidate mechanisms before more computationally intensive analyses are performed. Although the present investigation focuses on planar kinematic mechanisms and epicyclic geared mechanisms, the underlying graph-theoretic framework provides a strong foundation for extension to more complex mechanical systems. The proposed technique

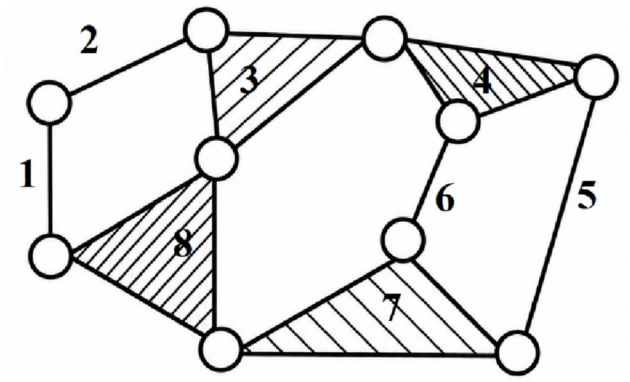


Fig. 12 Graph of Eight-links One-DOF PKC-2

Table 6 TDT for two PKCs with Six-links and One-degree of freedom

Fig. No.	Adjacency Matrix [A]	Total Distance (TDT)
Figure 11	$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 2 & 3 & 4 & 3 & 2 \\ 2 & 1 & 0 & 1 & 2 & 3 & 2 & 1 \\ 3 & 2 & 1 & 0 & 1 & 2 & 1 & 2 \\ 4 & 3 & 2 & 1 & 0 & 1 & 2 & 3 \\ 3 & 4 & 3 & 2 & 1 & 0 & 1 & 2 \\ 2 & 3 & 2 & 1 & 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 3 & 2 & 1 & 0 \end{bmatrix}$	112
Figure 12	$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 3 & 2 & 1 \\ 1 & 0 & 1 & 2 & 3 & 3 & 3 & 2 \\ 2 & 1 & 0 & 1 & 2 & 2 & 2 & 1 \\ 3 & 2 & 1 & 0 & 1 & 1 & 2 & 2 \\ 4 & 3 & 2 & 1 & 0 & 2 & 1 & 2 \\ 3 & 3 & 2 & 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 2 & 2 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 2 & 1 & 0 \end{bmatrix}$	106

therefore represents a practical contribution toward efficient structural synthesis and automated mechanism design.

Conclusion and Future Study

This study presents a novel technique based on Total Distance Technique for detecting isomorphism in planar kinematic mechanisms and epicyclic geared mechanisms using graph theory and adjacency matrix representation. The proposed method computes a unique total distance parameter through simple matrix operations, providing an efficient criterion for distinguishing structurally identical and non-identical mechanism graphs. Validation through several examples involving mechanisms with different numbers of links and degrees of freedom demonstrates that the proposed approach produces results consistent with well-established methods reported in the literature while offering a simpler computational procedure.

One of the major advantages of the proposed technique is its computational simplicity, making it suitable for rapid implementation in MATLAB and other computer-aided design environments. Compared with several existing graph-based approaches, the proposed method reduces mathematical complexity without compromising the accuracy of isomorphism identification, thereby improving the efficiency of structural synthesis and mechanism enumeration. Despite these encouraging results, several limitations should be acknowledged. The present work has been validated only for planar kinematic mechanisms and epicyclic geared mechanisms with relatively limited numbers of links and degrees of freedom. Furthermore, the current implementation relies on adjacency matrix representation and does not directly address three-dimensional spatial mechanisms or highly complex multi-body systems. Additional validation using large-scale benchmark databases would further establish the robustness and general applicability of the proposed technique.

The scalability of the Total Distance Technique provides significant opportunities for future research. Owing to its straightforward computational framework, the method can be extended to mechanisms with higher numbers of links, multiple degrees of freedom, and large-scale automated mechanism generation. Its low computational cost also makes it suitable for integration into computer-aided structural synthesis and optimization algorithms.

Future work will focus on extending the proposed methodology to spatial mechanisms, robotic manipulators, complex vehicle transmission systems, and aerospace mechanisms. Furthermore, integration with Artificial Intelligence (AI) and Machine Learning (ML) techniques for automatic graph classification, intelligent mechanism synthesis, design optimization, and CAD/CAM-based automated mechanism generation represents a promising direction. These developments will further enhance the applicability of the proposed Total Distance Technique in next-generation computational mechanism design and intelligent engineering systems.

Declarations

Conflict of interest The author states they did not receive money from competing sources or face situations that may have changed the research shown in this report.

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